DAA LAB ASSIGNMENT WEEK-7

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1)Quick Sort by taking three different pivots and Quick select

CODE:

```
#include <stdio.h>

void swap(int *a, int *b){
   int t = *a;
   *a = *b;
   *b = t;
}

int median(int a[], int l, int r){
   int m = (l + r) / 2;

   if(a[l] > a[m]) swap(&a[l], &a[m]);
   if(a[l] > a[r]) swap(&a[l], &a[r]);
   if(a[m] > a[r]) swap(&a[m], &a[r]);

   swap(&a[m], &a[r-1]);
   return a[r-1];
}
```

```
int partition(int a[], int l, int r){
    int pivot = median(a, l, r);

    int i = l;
    int j = r - 1;

    while(1){
        while(a[++i] < pivot);
        while(a[--j] > pivot);

        if(i < j)
            swap(&a[i], &a[j]);
        else
            break;
    }

    swap(&a[i], &a[r-1]);
    return i;
}

void quicksort(int a[], int l, int r){
    if(l < r){
        int p = partition(a, l, r);

        quicksort(a, l, p - 1);
        quicksort(a, p + 1, r);
```

```
    }
}

int quickselect(int a[], int l, int r, int k){
    if(l == r)
        return a[l];

    int p = partition(a, l, r);

    if(k == p)
        return a[p];
    else if(k < p)
        return quickselect(a, l, p - 1, k);
    else
        return quickselect(a, p + 1, r, k);
}

int main(){

    int n, k;

    printf("Enter number of elements: ");
    scanf("%d", &n);

    int a[n];
```

```
    printf("Enter elements:\n");
    for(int i = 0; i < n; i++)
        scanf("%d", &a[i]);

    quicksort(a, 0, n - 1);

    printf("Sorted array:\n");
    for(int i = 0; i < n; i++)
        printf("%d ", a[i]);

    printf("\nEnter k value for kth smallest element: ");
    scanf("%d", &k);

    printf("kth smallest element = %d", quickselect(a, 0, n - 1, k - 1));

    return 0;
}
```

OUTPUT:

```
santhan@Santhan-PC:~$ nano quickselect.c
santhan@Santhan-PC:~$ gcc quickselect.c -o quickselect
santhan@Santhan-PC:~$ ./quickselect
Enter number of elements: 5
Enter elements:
8
3
7
1
9
Sorted array:
1 3 7 8 9
Enter k value for kth smallest element: 3
kth smallest element = 7santhan@Santhan-PC:~$
```